

Kali Setup & Home Lab Foundations

Your lab = your growth

Every professional cybersecurity skill is practiced somewhere you control. Master this setup once and reuse it forever.

Hardware requirements (minimum)

16 GB RAM recommended (4 for Kali, 4-8 for modern targets)

Two CPU cores per VM; enable virtualization in BIOS

60+ GB disk (Kali 40GB + targets 20GB)

x86-64 host

VirtualBox setup

Install **VirtualBox** (virtualbox.org) + **Extension Pack** for USB/VRDE

Download the **Kali VirtualBox image** (kali.org) – a ready OVA (easier than ISO)

Import: File → Import Appliance

Kali VM essentials

RAM 4096 MB, CPU 2, 40GB disk (dynamic)

Storage: SATA controller

Networking: we'll use host-only + NAT (see below)

After boot: run `sudo apt update && sudo apt full-upgrade`

Enable the non-root user in the pre-created kali user or create yours

Network topology

```
[ Host (VirtualBox) ]  
| Host-Only Network vboxnet0 (10.0.2.15 lab subnet)  
+--- Kali (10.0.0.10)  
+--- Target (10.0.0.20) Metasploitable/DVWA  
+--- Win10 (10.0.0.30) target for lateral labs  
NAT -> internet (updates, downloads)
```

NAT: Kali talks out to internet, targets unreachable from Kali

Host-only: Kali+targets all on lab subnet, isolated from your LAN

Bridged: risky (exposes on YOUR Wi-Fi) – avoid for beginners

Combine: give Kali TWO NICs (NAT + host-only) so you update but test locally

Snapshots are your superpower

After Kali setup → snapshot "clean-1"

After each exploit experiment → snapshot before/after

Roll back in seconds, never fear breakage

Tool hygiene

`sudo apt update` weekly

`sudo kali-linux-everything` only if you want EVERYTHING (600+ tools)

Prefer category metas: `sudo apt install metasploit-framework burpsuite wireguard`

Bookmark the Kali docs (kali.org/docs)

Rule of the lab: everything destructive happens inside VirtualBox on a snapshot-backed VM. Your host stays clean and your experiments stay reversible.

[← Module start](#)

[Information Gathering Tools →](#)